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LDS-Net: A Lightweight Dual-Branch Network for Slender Tree Branch Segmentation in Complex Natural Scenes

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Abstract

Accurate semantic segmentation of slender curvilinear structures, such as tree branches, in complex natural scenes remains challenging. The main difficulties arise from frequent occlusions, ambiguous boundaries, and limited edge computing resources. To address these issues, we propose LDS-Net, a lightweight dual-branch network designed for thin and continuous branch structures. The Detail-Aware Branch uses the Dynamic Snake Convolution (DSConv) to model irregular local geometry, while the Context-Aware Branch uses a Spatial Efficient Separable Pyramid module (SESP) to capture multi-scale context. To improve segmentation under occlusion and boundary ambiguity, we further integrate a Global Topology Transformer module (GTT) and a boundary guidance mechanism (BG). These features are fused via a Pixel-Wise Attention Fusion module (PAF) and optimized using a multi-head compound loss. Experiments on USTD and N-ABSD show that LDS-Net outperforms six representative networks. It also requires only 14.19 G FLOPs, a 24% reduction compared with PIDNet-s. These results suggest that LDS-Net has potential for future deployment on resource-constrained agricultural and ecological monitoring platforms.

Keywords: semantic segmentation; lightweight dual-branch network; feature extraction; tree branch; deep learning

1. Introduction

Accurate ecosystem monitoring and management have become important research topics in environmental science, urban planning, and modern agriculture. As a crucial component of terrestrial ecosystems, trees [1] play an important role in maintaining ecological balance, regulating urban climate, and supporting agricultural and forestry production. Therefore, accurate acquisition and analysis of tree structural characteristics [2] are essential for ecological assessment, smart city development, and precision agriculture and forestry.

Among various tree components, branches [3] are both informative and difficult to perceive visually. Their spatial distribution and continuity provide important cues for evaluating tree growth and health. Quantitative branch characteristics, such as thickness, branching patterns, and geometric arrangement, support a wide range of downstream applications. These applications include urban tree health assessment, management of valuable tree species, and roadside safety inspection [4]. In agricultural and orchard environments, branch structures also affect precision pruning, spray path planning, and autonomous navigation for harvesting robots [5]. For automatic dormant pruning, Sun et al. [6] developed a real-time method to detect and characterize pear tree trunks and upright branches,

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demonstrating the practical need for robust tree-component perception in orchard robots. Albaroudi et al. [7] estimated branch geometry and hierarchical relationships from RGB images to support robotic pruning decisions.

However, obtaining high-precision branch structure information remains challenging. Traditional methods often rely on manual measurements with tools such as calipers and protractors. These methods are time-consuming, labor-intensive, and may involve safety risks. To reduce manual effort, many automatic branch perception methods have been developed over the past decades. Early studies mainly used conventional image processing techniques, including thresholding [8], edge-based segmentation [9], region growing [10], graph-based methods [11], watershed algorithms [12], and K-means clustering [13]. However, these methods usually require high-quality images, and their accuracy decreases sharply when environmental conditions change.

To obtain comprehensive 3D structural information, point-cloud-based sensing and analysis methods have been widely used [14]. These methods include Terrestrial Laser Scanning (TLS) [15], Mobile Laser Scanning (MLS) and Unmanned Aerial Scanning (ULS) [16]. They provide detailed 3D data for analyzing branch morphology and canopy structure. For instance, Peng et al. [17] systematically evaluated an automated branch angle measurement method based on TLS point clouds. Yrttimaa et al. [18] used TLS point clouds and coordinate transformation to separate branches from trunks. In complex urban scenes, Li et al. [19] proposed an individual street tree extraction method based on MLS point clouds and non-photosynthetic component clustering. Wu et al. [20] further introduced a coarse registration method for TLS-ULS point clouds using line segment feature descriptors. However, as noted by Hartley et al. [21], although laser scanning is useful for branch segmentation, research gaps persist in fine-branch structural analysis. Furthermore, the high instrument cost and operational complexity of direct 3D scanning systems limit their widespread application.

To address cost and efficiency challenges, researchers have increasingly turned to visual methods based on 2D imagery. Object detection methods primarily use rectangular bounding boxes for coarse localization of targets. Pagire et al. [22] reviewed the evolution of deep learning-based object detectors in several domains, including agriculture and remote sensing. For agricultural perception, many studies have optimized standard detection architectures for specific tree species or structures. For instance, Akdoğan et al. [23] introduced PP-YOLO with histogram and wavelet preprocessing to locate apple and cherry trees in orchards. For large-scale canopy mapping, Hnida et al. [24] presented an enhanced multi-scale architecture that combines CSPNet with a Feature Pyramid Network to detect olive tree crowns from UAV orthophotos. In addition to bounding-box localization, dual-task learning frameworks have also been explored. Rahman et al. [25] used a hybrid self-attention U-Net framework to jointly predict segmentation masks and centroid heatmaps for dead tree detection. For curvilinear sub-structure perception, Lin et al. [26] integrated YOLO object detection with Semi-Global Block Matching stereo vision for autonomous drone-based radiata pine pruning applications.

Although these detection-based methods have achieved promising results, they still face structural limitations when applied to branch perception. Rectangular bounding boxes often include substantial background noise and cannot accurately represent the irregular, slender, and continuous boundaries of natural tree branches. Cascading a detection front-end with stereo vision or segmentation modules [25,26] can improve boundary alignment, but such multi-stage systems also increase computational cost. This makes efficient deployment on agricultural edge devices more difficult.

Semantic segmentation provides a more suitable solution for precise pixel-level understanding of slender branch structures. Early deep learning-based segmentation studies

mainly relied on single-backbone encoder–decoder architectures. Shang et al. [27] incorporated residual learning into U-Net-based networks to improve segmentation accuracy in orchard environments. Peng et al. [28] proposed a ResDense-focal-DeepLabV3+ [29] model for litchi branch semantic segmentation in robotic harvesting. Sun et al. [30] combined stereo vision and semantic segmentation for rapid tree trunk localization in pear orchards. PSPNet [31] is a representative pyramid-based framework for enhancing semantic feature representation, while Chae et al. [32] proposed TinySeg for image segmentation on tiny embedded systems.

Recent work in digital agriculture and forestry has increasingly used multi-branch architectures and multi-scale attention mechanisms to combine complementary feature representations. In 3D data analysis, Cao et al. [33] proposed an adaptive cuboid regional growth algorithm for high-precision trunk-branch segmentation. In 2D image analysis, Liu et al. [34] developed MMCAlexNet for tree species classification by integrating a multi-branch convolutional module and a mixed-domain attention module. Olzhayev et al. [35] introduced a YOLO-like multi-branch deep learning architecture for localized target parsing. Other multi-branch methods, such as HAFNet [36], MDASN [37], and the residual-network-based method of Qu et al. [38], have also shown the value of combining global and local feature branches. Bu et al. [39] proposed a CNN–Transformer dual-branch collaborative network for farmland segmentation, showing that local spatial details and global semantic context can be jointly used for complex high-resolution agricultural segmentation.

To further address occlusion, long-range structural dependencies, and ambiguous boundaries, convolutional branches have been combined with Transformer-based reasoning or boundary refinement modules. Deng et al. [40] designed a spatial-frequency domain multi-branch decoder for real-time semantic segmentation. To address semantic discontinuities, DMSF-Net [41] integrates deformable convolution and attention gating for efficient medical image segmentation. For lightweight integration, Wan et al. [42] developed U2ESPNet, while Zeng et al. [43] introduced LED-Net. Shan et al. [44] proposed a multi-scale boundary-aware network for remote sensing semantic segmentation, in which uncertain boundary regions are explicitly modeled to improve feature representation in complex high-resolution scenes. More recently, Ren et al. [45] introduced a quantum-enhanced U-Net that combines a hybrid quantum–classical architecture with a convolutional block attention module to reduce boundary fragmentation in canopy segmentation.

Despite these advances, extracting slender and curvilinear branches in complex natural scenes remains challenging. First, branches exhibit extreme aspect ratios and irregular topological variations. Standard residual CNN backbones, such as ResNet [46], use fixed rectangular convolutional kernels. These kernels may introduce background noise and cause structural disconnections when modeling thin and curved branches. Second, orchard and street scenes present intense illumination variations and dense occlusions, exacerbating ambiguous foreground–background boundaries. Third, branch pixels occupy only a small proportion of the image, resulting in severe class imbalance. Some methods address these issues by stacking heavy convolutional layers, introducing complex Transformer [47] modules, or using more complex mechanisms. However, these designs incur high computational costs, making them impractical for deployment on resource-constrained edge devices. Conversely, overly lightweight networks may fail to preserve the structural integrity of slender branches. This limitation highlights the need for a segmentation network that can effectively balance detail information, global context modeling, and computational efficiency.

To address this accuracy–efficiency trade-off, we propose LDS-Net, a lightweight dual-branch semantic segmentation network tailored for slender tree branch segmentation in complex natural scenes. Inspired by PIDNet-s [48], LDS-Net uses a detail-context dual-branch architecture. However, branch segmentation is not only a pixel-level clas-

sification task. It is also a structure-sensitive problem involving curvilinear continuity, occlusion-induced ambiguity, boundary uncertainty, foreground–background imbalance, and computational efficiency. Unlike PIDNet-s, which introduces an additional boundary branch, LDS-Net derives boundary-sensitive cues from the discrepancy between the Detail-Aware and Context-Aware Branches. Compared with related dual-branch segmentation frameworks, such as HAFNet [36], MDASN [37], Qu et al. [38], and the CNN–Transformer dual-branch collaborative network proposed by Bu et al. [39], LDS-Net further treats cross-branch feature discrepancies as a source of boundary-sensitive information. In this way, boundary modeling is coupled with dual-branch feature collaboration rather than being introduced as an isolated auxiliary refinement.

The principal contributions of this paper are summarized as follows:

1. We propose a task-specific lightweight network for slender tree branch segmentation. LDS-Net is designed for branch structures affected by background interference, illumination variation, and leaf occlusion. It jointly considers local curvilinear continuity, contextual reasoning, boundary discrimination, foreground–background imbalance, and computational efficiency.
2. We design a detail–context dual-branch backbone. The Detail-Aware Branch employs the Dynamic Snake Hybrid Residual Block (DSHRB) with Dynamic Snake Convolution (DSConv) [49] to adapt to irregular branch trajectories and preserve local structural continuity. The Context-Aware Branch uses the Multi-Scale Efficient Residual Block (MERB) with the Spatial Efficient Separable Pyramid module (SESP) to capture multi-scale semantic context with low computational cost.
3. We introduce a cross-branch boundary cue generation strategy. The proposed boundary guidance mechanism (BG) extracts boundary-sensitive cues from the feature discrepancy between the Detail-Aware and Context-Aware Branches. This design avoids adding an independent boundary branch and makes boundary modeling part of the dual-branch feature collaboration.
4. We incorporate occlusion-aware structural reasoning and pixel-wise adaptive fusion. The Global Topology Transformer module (GTT) is applied to high-level features to enhance structural reasoning for partially occluded branches. The Pixel-Wise Attention Fusion module (PAF) adaptively integrates detail, context, and boundary cues at each spatial location.
5. We formulate a multi-head compound loss for slender-structure supervision. The proposed loss provides complementary supervision for the main segmentation output and auxiliary predictions, aiming to improve structural connectivity, boundary discrimination, and foreground–background balance.
6. We conduct experiments on USTD and N-ABSD to evaluate the effectiveness and efficiency of LDS-Net. The results show that LDS-Net achieves a favorable accuracy–efficiency trade-off compared with representative semantic segmentation networks. Ablation studies further verify the effectiveness of the dual-branch backbone, boundary guidance strategy, attention-based fusion, and multi-head supervision.

2. Materials and Methods

2.1. Overall Architecture

Standard single-backbone networks often lose fine spatial details when segmenting slender and curvilinear branches. To address this problem, LDS-Net adopts a lightweight dual-branch architecture, as shown in Figure 1. After a shared Conv Stem, the extracted features are fed into two parallel branches. The Detail-Aware Branch (B_{detail}) is designed to preserve thin branch structures and local continuity. In this branch, the Dynamic Snake Hybrid Residual Block (DSHRB) with the Dynamic Snake Convolution (DSConv) helps

the network adapt to irregular branch shapes and reduces discontinuities caused by fixed convolutional kernels, as described in Section 2.2. The Context-Aware Branch (B_{context}) is designed to capture broader contextual information. In this branch, the Multi-Scale Efficient Residual Block (MERB) aggregates multi-scale features with low computational cost. The Global Topology Transformer module (GTT) further uses the window-based self-attention (W-MSA) to capture structural relationships and the offset prediction (OP) to improve the continuity of occluded branches, as described in Section 2.3.

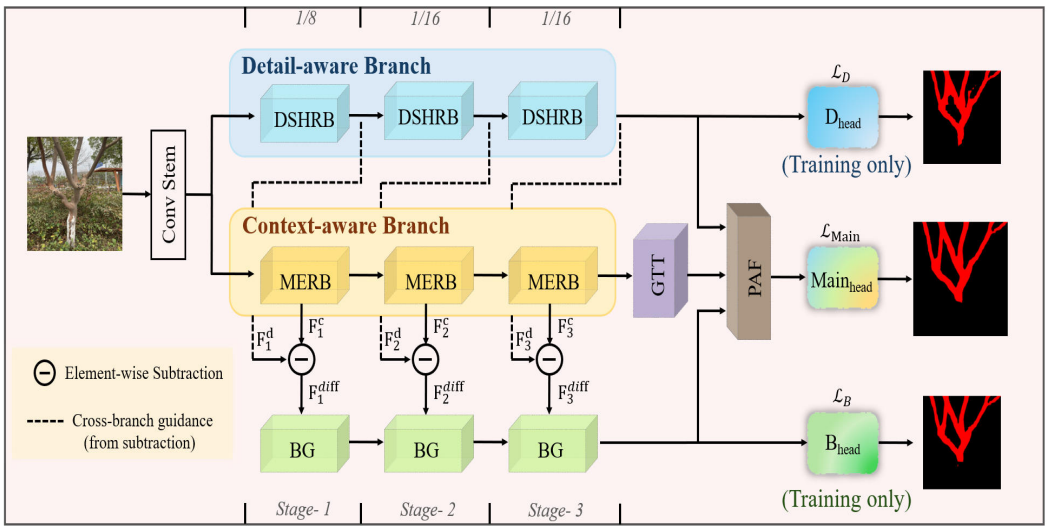


Figure 1. Overview of LDS-Net architecture. The network contains a shared Conv Stem, a Detail-Aware Branch, a Context-Aware Branch, BG, GTT, PAF, and three prediction heads. The D_{head} and B_{head} are auxiliary heads used only during training.

To improve foreground–background boundary discrimination, the boundary guidance mechanism (BG) uses the feature difference between the two branches. At the j-th stage, features from the Detail-Aware Branch and the Context-Aware Branch are denoted as $F_j^d, F_j^c \in \mathbb{R}^{C_j \times H_j \times W_j}$. The cross-branch difference feature is computed as

$$F_j^{\text{diff}} = F_j^d - F_j^c, \quad j \in \{1, 2, 3\}. \tag{1}$$

This difference feature highlights areas where detail and context responses are inconsistent. These areas usually correspond to branch regions with unclear boundaries or background interference. Before fusion, the difference features are refined by a residual bottleneck unit to reduce noise.

Finally, the Pixel-Wise Attention Fusion module (PAF) integrates the detail features, context features, and boundary-guided features, as described in Section 2.4. Since different image regions require different cues, PAF assigns pixel-wise weights to these features before fusion. This allows the network to use more local detail information near thin branches, more contextual information in occluded regions, and more boundary information around unclear edges. During training, the multi-head compound loss supervises the main output together with the detail-aware and boundary-aware auxiliary outputs, as described in Section 2.5.

Overall, LDS-Net combines local detail preservation from the Detail-Aware Branch, multi-scale contextual modeling from the Context-Aware Branch, structural reasoning from GTT, boundary cue extraction from BG, adaptive feature fusion from PAF, and complementary supervision from the multi-head compound loss. These components jointly improve slender branch segmentation in complex natural scenes.

2.2. Detail-Aware Branch

To capture fine branch structures and local continuity, we design the Detail-Aware Branch (B_{detail}), which uses DSHRB as its basic building block. As illustrated in Figure 2a, DSHRB contains two parallel paths: a static multi-scale dilated convolutional path (P_{sta}) and a dynamic snake sampling path (P_{dyn}). The static path provides stable multi-scale local features, while the dynamic path adapts the sampling positions to irregular branch shapes. Together, these two paths help reduce structural discontinuities and aliasing effects caused by fixed convolutional kernels when modeling slender and curved branch structures.

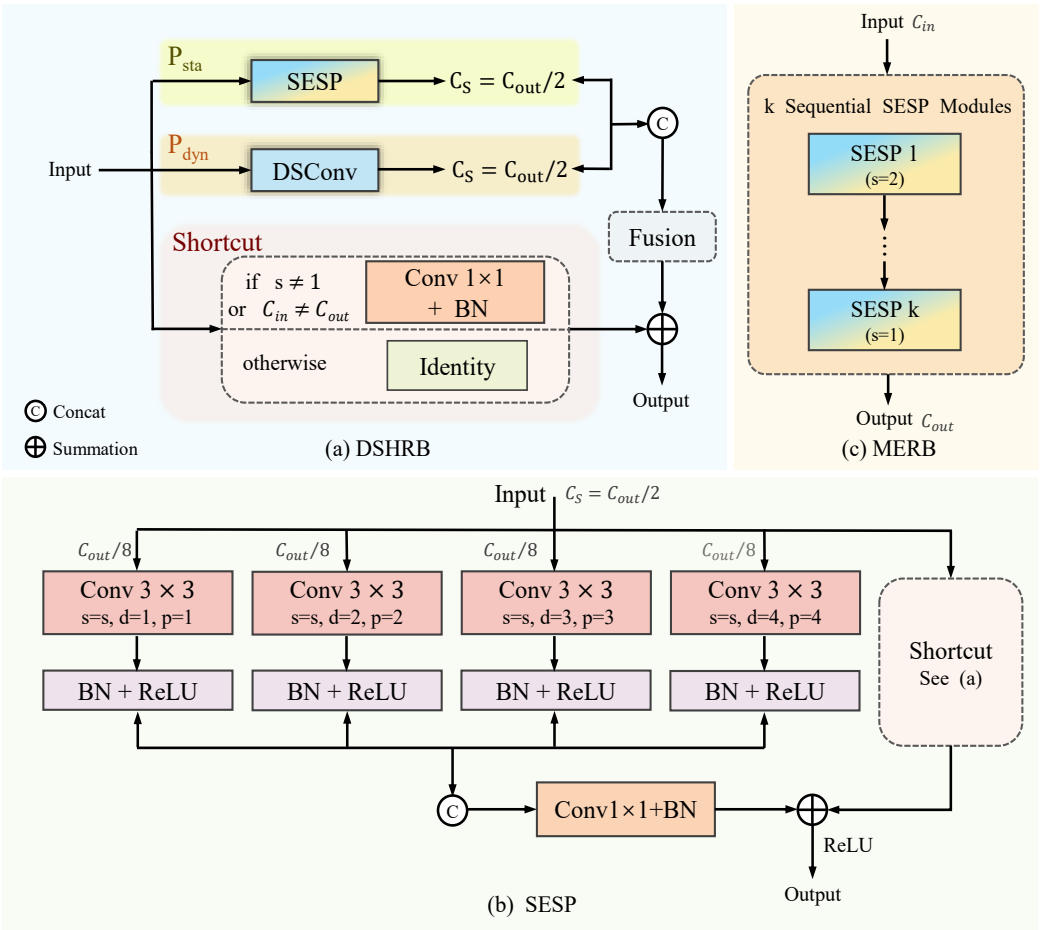


Figure 2. Dual-branch backbone modules. (**a**) DSHRB contains a static path, a dynamic path, and a shortcut. (**b**) SESP uses four parallel dilated convolution branches with dilation rates $d \in \{1, 2, 3, 4\}$. (**c**) MERB stacks k sequential SESP modules.

- **Static Multi-Scale Dilated Convolutional Path.**
  This path employs SESP to enlarge the receptive field with low computational cost, as illustrated in Figure 2b. Given an input feature map $X \in \mathbb{R}^{B \times C_{\text{in}} \times H \times W}$, SESP in DSHRB outputs $C_{\text{s}} = C_{\text{out}}/2$ channels, where C_{out} denotes the channel width of the whole DSHRB. The output is divided into four parallel channel sub-branches, each with $C_{\text{s}}/4 = C_{\text{out}}/8$ channels. Each sub-branch is processed by a 3×3 convolution with stride s, dilation rate $d_l \in \{1, 2, 3, 4\}$, and padding $p_l = d_l$ to maintain spatial alignment. The four outputs are concatenated along the channel dimension and fused by a 1×1 convolution followed by batch normalization. When $C_{\text{in}} \neq C_{\text{s}}$ or $s \neq 1$, a shortcut branch with a 1×1 convolution and batch normalization is used for feature alignment; otherwise, an identity mapping is adopted.

- **Dynamic Snake Sampling Path.**
  The dynamic snake sampling path incorporates DSConv to process the remaining feature channels. Unlike standard convolutions, which are constrained by a fixed rectangular grid $\mathcal{R} \in \{(-1,-1),\ldots,(1,1)\}$, DSConv introduces position-dependent offsets to adapt the convolutional kernel to branch orientations. Through an iterative coordinate offset strategy, the receptive field can deform along the curvilinear trajectories of tree branches. This helps reduce sampling deviations of conventional kernels and improves the continuity of slender branch structures in complex backgrounds. Since the predicted coordinates are usually non-integer values, bilinear interpolation is used to resample features and ensure precise responses at the corresponding spatial locations.

2.3. Context-Aware Branch

2.3.1. MERB: Multi-Scale Efficient Residual Block

Unlike B_{detail}, which focuses on local details, B_{context} aims to capture broader contextual information. To achieve this with low computational cost, B_{context} uses MERB as its fundamental building block, as illustrated in Figure 2c. Each MERB consists of k sequential SESP modules. The first SESP employs a stride of $s = 2$ for spatial downsampling and maps the channel dimension from C_{in} to C_{out}. The remaining $(k-1)$ SESP modules use a stride of $s = 1$ to further aggregate multi-scale contextual information while maintaining the spatial resolution.

2.3.2. GTT: Global Topology Transformer

To improve the modeling of occluded branches, we introduce GTT on the high-level semantic features of B_{context}, as illustrated in Figure 3a. GTT adopts a pre-norm residual architecture comprising three sequential sub-modules: Window-Based Multi-head Self-Attention (W-MSA), Offset Prediction (OP), and Feed-Forward Network (FFN). W-MSA captures structural relationships within local windows with reduced computational cost. OP enhances directional continuity through offset prediction, and FFN promotes channel-wise feature interaction.

- **W-MSA:** To reduce the computational cost of self-attention, W-MSA partitions the input feature map into non-overlapping $M \times M$ windows (we set $M = 7$) and performs multi-head self-attention within each window. Given an input feature map $X \in \mathbb{R}^{B\times C\times H\times W}$, each window is flattened into a token sequence $X_w \in \mathbb{R}^{N\times \dim}$, where $N = M \times M$. Queries, keys, and values are generated by linear projections and divided into h attention heads. For the i-th attention head, $Q_i, K_i, V_i \in \mathbb{R}^{N\times d_h}$, where $d_h = C/h$. The attention computation is defined as

$$\text{Attention}_i(Q,K,V) = \text{Softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_h}}\right) V_i, \tag{2}$$

  where $Q_i K_i^\top$ denotes the token similarity matrix within a window. The outputs of all attention heads are concatenated and projected back to the feature space. Finally, the window features are rearranged to the original spatial layout, as illustrated in Figure 3b.
- **OP:** To improve the continuity and directional consistency of slender branch structures, OP first employs a 3×3 depthwise convolution to extract local structural features, followed by a 1×1 pointwise convolution for channel aggregation. Batch normalization and ReLU activation are then applied to obtain an intermediate feature map

$\hat{X}$. Subsequently, $\hat{X}$ is fed into a 1×1 offset convolution to predict two-dimensional structural offsets:

$$\Delta = (\Delta_x, \Delta_y) \in \mathbb{R}^{2 \times H \times W}. \tag{3}$$

To avoid unstable large offsets, the predicted offsets are constrained by the $\tanh(\cdot)$ function and applied to the residual features in a gated manner:

$$X_{\text{op}} = \hat{X} + X \odot \tanh(\Delta_x + \Delta_y), \tag{4}$$

where $\odot$ denotes element-wise multiplication. The term $\tanh(\Delta_x + \Delta_y)$ serves as a single-channel spatial gate and is broadcast along the channel dimension, as illustrated in Figure 3c.

- **FFN:** To facilitate channel-wise information exchange, FFN contains two 1×1 convolutions. The first expands the channel dimension to $4C$, and the second projects it back to C. GELU activation is applied between the two convolutions, and dropout is used to improve generalization, as illustrated in Figure 3d.

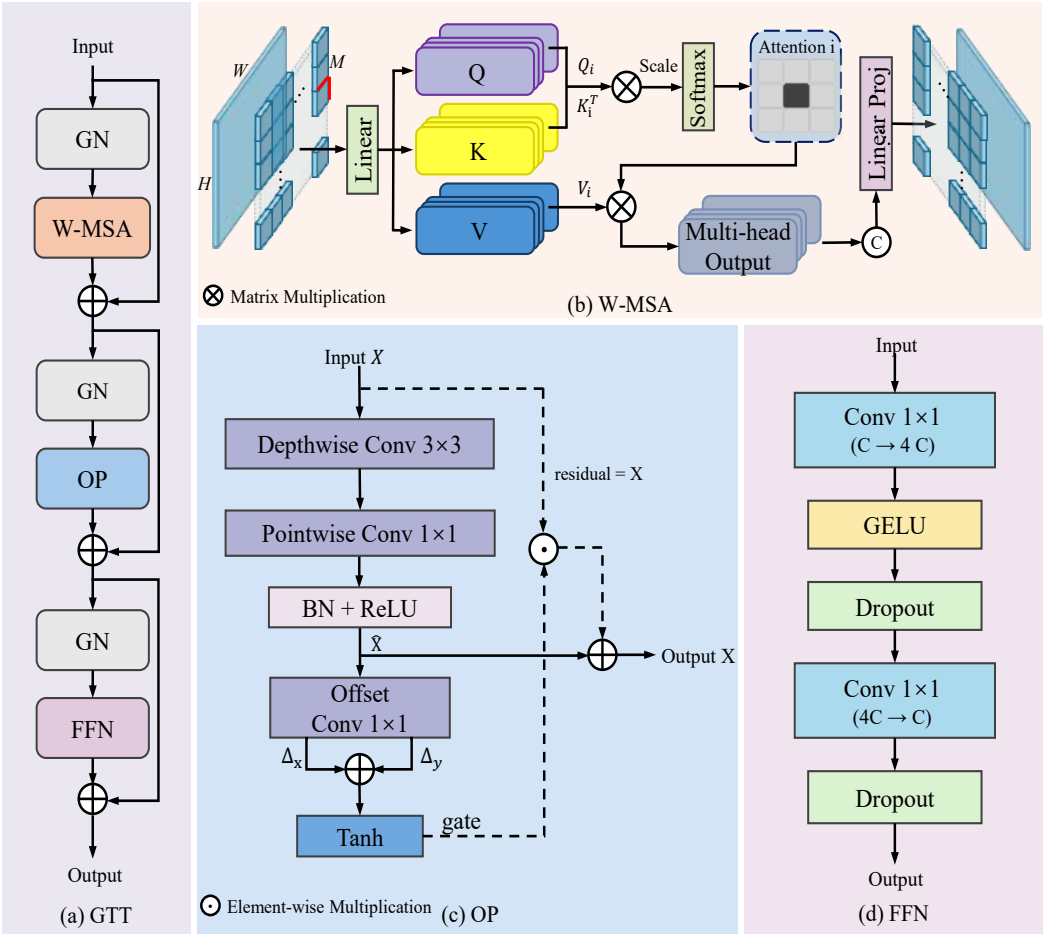


Figure 3. Structure of the GTT module. (**a**) GTT contains three pre-norm residual sub-modules. (**b**) W-MSA performs self-attention within non-overlapping windows. (**c**) OP predicts structural offsets to improve branch continuity and the dashed arrows indicate the offset-derived spatial gating applied to the residual features. (**d**) FFN performs channel-wise feature interaction.

2.4. PAF: Pixel-Wise Attention Fusion

Given the detail feature X_d, context feature X_c, and boundary-guided feature X_b, the Pixel-Wise Attention Fusion module (PAF) combines these three types of features with pixel-wise weights, as illustrated in Figure 4. Different image regions may rely on different

cues. Therefore, PAF adjusts the contribution of detail, context, and boundary-guided features before fusion.

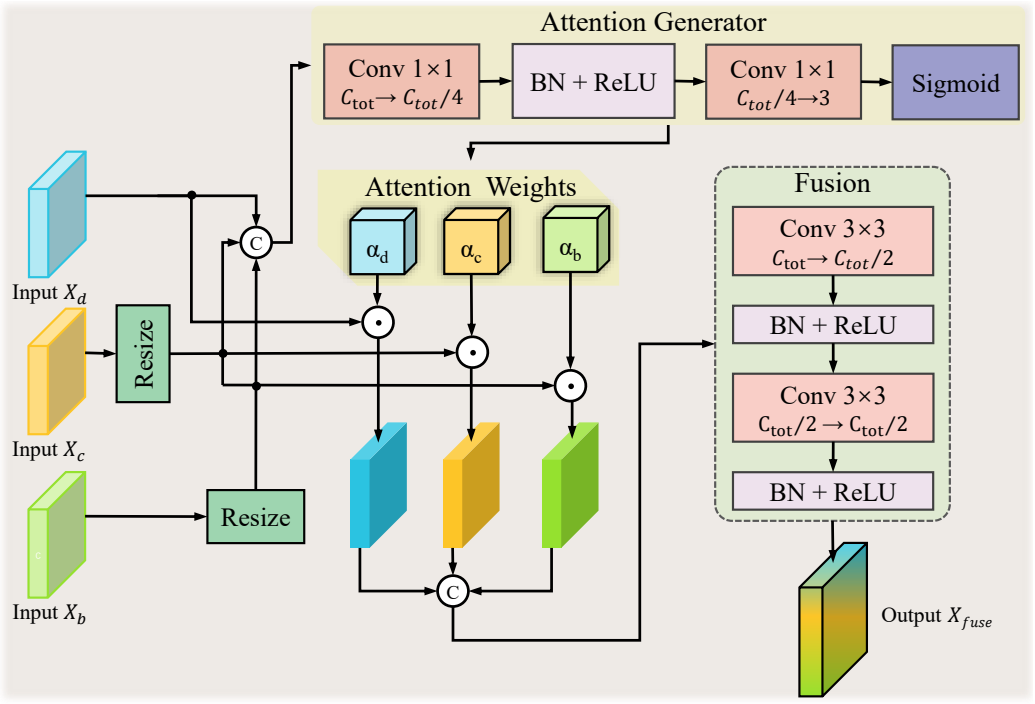


Figure 4. Pixel-Wise Attention Fusion module. PAF predicts pixel-wise weights $(\alpha_d, \alpha_c, \alpha_b)$ for detail, context, and boundary-guided features, and then fuses the weighted features to obtain the final representation X_{fuse}.

Features from different branches may have different spatial sizes. Thus, X_c and X_b are first resized to the same spatial size as X_d using bilinear interpolation. The aligned features are denoted as $\bar{X}_d$, $\bar{X}_c$, and $\bar{X}_b$. They are then concatenated along the channel dimension to form X_{cat}, whose channel number is $C_{\text{tot}} = C_d + C_c + C_b$.

To generate pixel-wise weights, X_{cat} is fed into an attention generator composed of two 1×1 convolutions:

$$A = \sigma\Big(\text{Conv}^{(2)}_{1\times1}\Big(\delta\Big(\text{BN}\Big(\text{Conv}^{(1)}_{1\times1}(X_{\text{cat}})\Big)\Big)\Big)\Big), \tag{5}$$

where $\text{Conv}^{(1)}_{1\times1}$ reduces the channel dimension from C_{tot} to $C_{\text{tot}}/4$, and $\text{Conv}^{(2)}_{1\times1}$ maps features to 3 channels, $\delta(\cdot)$ denotes ReLU, and $\sigma(\cdot)$ denotes the sigmoid function. The attention map $A \in \mathbb{R}^{3\times H\times W}$ is partitioned into three single-channel masks:

$$A = [\alpha_d; \alpha_c; \alpha_b], \quad \alpha_k \in \mathbb{R}^{1\times H\times W},\ k \in \{d, c, b\}. \tag{6}$$

The three masks control the contributions of detail, context, and boundary information at each pixel.

Each aligned feature is then weighted by its corresponding mask:

$$\tilde{X}_k = \bar{X}_k \odot \alpha_k, \quad k \in \{d, c, b\}, \tag{7}$$

Finally, the weighted features are concatenated and fused by a lightweight fusion sub-network to obtain the final fused representation. In this way, PAF can emphasize local details near thin branches, contextual cues in occluded regions, and boundary cues around unclear edges.

2.5. Loss Function

To address foreground–background imbalance, slender branch structures, and ambiguous boundaries, LDS-Net uses a multi-head compound loss. After PAF, the network produces three segmentation predictions: the main prediction from $\text{Main}_{\text{head}}$, the detail-aware auxiliary prediction from D_{head}, and the boundary-guided auxiliary prediction from B_{head}. The two auxiliary heads are used only during training. They provide additional supervision for detail and boundary learning. During inference, only the $\text{Main}_{\text{head}}$ is retained. The compound loss includes Focal Loss, Dice Loss, and Boundary-Weighted Cross-Entropy Loss.

- **Focal Loss ($\mathcal{L}_{\text{focal}}$):** Focal Loss is used to reduce the effect of foreground–background class imbalance. Given the ground-truth label y_m, predicted logit z_m, and predicted probability p_{m,y_m} for the m-th valid pixel, the loss is defined as

$$\mathcal{L}_{\text{focal}} = \frac{1}{|\Omega|} \sum_{m \in \Omega} \alpha_{y_m} \left(1 - p_{m,y_m}\right)^{\gamma} \text{CE}(z_m, y_m), \tag{8}$$

  where $|\Omega|$ denotes the number of valid pixels, γ is the focusing parameter, and α_{y_m} is the class weighting factor. We set $\gamma = 3.0$. Focal Loss is applied to $\text{Main}_{\text{head}}$, D_{head}, and B_{head}.
- **Dice Loss ($\mathcal{L}_{\text{dice}}$):** Dice Loss is used to improve spatial consistency for small branch regions. It is defined as

$$\mathcal{L}_{\text{dice}} = \frac{1}{C} \sum_{c=1}^{C} \left(1 - \frac{2 \sum_{m \in \Omega} p_{m,c}\, y_{m,c} + \epsilon}{\sum_{m \in \Omega} p_{m,c} + \sum_{m \in \Omega} y_{m,c} + \epsilon}\right), \tag{9}$$

  where C is the number of classes, and ϵ is a smoothing term. We set $\epsilon = 1.0$. To use Dice supervision from different prediction heads, the multi-head Dice Loss is defined as

$$\mathcal{L}_{\text{dice}}^{\text{ms}} = \frac{\mathcal{L}_{\text{dice}}^{\text{main}} + 0.5\, \mathcal{L}_{\text{dice}}^{d} + 0.3\, \mathcal{L}_{\text{dice}}^{b}}{1.8}, \tag{10}$$

  where the denominator normalizes the multi-head Dice Loss.
- **Boundary-Weighted Cross-Entropy Loss ($\mathcal{L}_{\text{bc}}$):** Boundary-Weighted Cross-Entropy Loss is used to strengthen supervision on branch boundaries. A binary boundary mask $M_{\text{bc}}^{(m)} \in \{0, 1\}$ is generated from the ground-truth labels by computing first-order differences in the horizontal and vertical directions. The loss is defined as

$$\mathcal{L}_{\text{bc}} = \frac{1}{|\Omega|} \sum_{m \in \Omega} \left(1 + \theta\, M_{\text{bc}}^{(m)}\right) \text{CE}(z_m, y_m), \tag{11}$$

  where θ is the boundary weighting coefficient. We set $\theta = 2.0$. Thus, boundary pixels receive larger loss weights. This loss is applied only to the output of $\text{Main}_{\text{head}}$.

The main head is supervised by Focal Loss, multi-head Dice Loss, and Boundary-Weighted Cross-Entropy Loss. The two auxiliary heads are supervised by Focal Loss. Thus, the total training loss is defined as

$$\mathcal{L}_{\text{total}} = \lambda_{\text{Main}} \left(\mathcal{L}_{\text{focal}}^{\text{Main}} + \lambda_{\text{dice}} \mathcal{L}_{\text{dice}}^{\text{ms}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}}\right) + \lambda_D \mathcal{L}_{\text{focal}}^{D} + \lambda_B \mathcal{L}_{\text{focal}}^{B}, \tag{12}$$

where $\lambda_{\text{dice}} = 0.5$, $\lambda_{\text{bc}} = 0.2$, $\lambda_{\text{Main}} = 1.0$, $\lambda_D = 0.7$, and $\lambda_B = 0.5$.

3. Results

The proposed network is trained and evaluated on two representative open branch segmentation datasets: the Urban Street Tree Dataset (USTD) [50] and the NWPU-Apple Branch Segmentation Dataset (N-ABSD) [51].

3.1. Datasets

- **Urban Street Tree Dataset (USTD).** USTD is a publicly available, comprehensive tree dataset covering 50 tree species from 10 urban street scenes. The images were captured under varying lighting conditions, seasons and shooting angles with tree parts finely annotated using polygons. The dataset comprises a total of 41,467 high-resolution images. It provides 22,872 finely annotated images suitable for image classification and instance segmentation tasks. The dataset comprises multiple classification and segmentation sub-datasets, including leaves, trunks and branches. We selected the branch sub-dataset, which covers tree species from 13 cities. It contains 1485 high-resolution RGB images and corresponding annotations. It should be noted that we focus solely on the semantic segmentation task of branches and background, without distinguishing between different types of branches. Consequently, the original category labels were merged and remapped, and the dataset was divided into training, validation, and test sets at an 8:1:1 ratio.
- **NWPU-Apple Branch Segmentation Dataset (N-ABSD).** N-ABSD is an open dataset constructed by the Northwest Polytechnic University research team for the task of semantic segmentation of apple tree branches. The experimental images were collected from an apple orchard in Liquan County, Shaanxi Province, China. The orchard employs a planting layout with a row spacing of 3.5 m and a plant spacing of 1.2 m. Data collection took place between September and October 2024, from 08:00 to 17:00, using a ZED camera and a laptop to capture images at a resolution of 1280 $\times$ 720 pixels. To closely replicate the operational environment of orchard harvesting robots, the camera was positioned approximately 50–80 cm from the tree branches. The dataset covers scenes of fruit tree branches across different time periods, lighting conditions and shooting angles. A total of over 2000 images were collected. In total, 1486 valid images were retained and divided into training, validation and test sets in a 7:2:1 ratio.

3.2. Implementation Details and Training Strategy

3.2.1. Implementation Details

All experiments were conducted on the Windows operating system, using Python 3.8.20 and the PyTorch 2.4.1 deep learning framework for model training and inference. During the experiments, training and inference were accelerated with CUDA 12.4 and cuDNN 9.1.0. All experiments were conducted on an NVIDIA RTX 4090 GPU (24 GB) using Automatic Mixed Precision (AMP). All models were optimized under a unified data preprocessing workflow and training configuration to ensure the fairness and reproducibility of the experimental results.

3.2.2. Training Strategy

Regarding data processing, all input images were uniformly resized to a resolution of 512 $\times$ 512, and data loading and augmentation were implemented using the Albumentations library. During the training phase, augmentation strategies such as random flipping, 90° rotation, affine transformation, and color dithering were employed, whereas only resizing and normalization were used during validation and testing. The network was trained with the AdamW optimizer, using an initial learning rate of 1×10^{-3} and a weight

decay of 1×10^{-4}. A polynomial learning rate decay strategy (power = 0.9) was adopted. The maximum number of training epochs was set to 120, with a batch size of 32.

3.3. Evaluation Metrics

We evaluate LDS-Net from two aspects: segmentation accuracy and computational cost.

3.3.1. Accuracy Metrics

We adopt several widely used evaluation metrics in the field of semantic segmentation, including Intersection over Union (IoU), mean Intersection over Union (mIoU), Pixel Accuracy (Accuracy), mean Pixel Accuracy (mPA), and mean F1-score (mF1). These metrics are calculated based on the confusion matrix, involving the counts of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). The specific definitions are as follows:

- **Intersection over Union (IoU):** Measures the overlap between the predicted segmentation and the ground truth for a specific class c:

$$\mathrm{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{13}$$

- **mean Intersection over Union (mIoU):** Represents the average of IoU values across all C classes, serving as the core metric for overall segmentation performance:

$$\mathrm{mIoU} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{IoU}_c \tag{14}$$

- **Pixel Accuracy (Accuracy):** Indicates the proportion of correctly classified pixels out of the total pixels in the image:

$$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

- **mean Pixel Accuracy (mPA):** Calculates the average of pixel accuracy for each class, providing a balanced assessment especially when class imbalance exists:

$$\mathrm{mPA} = \frac{1}{C} \sum_{c=1}^{C} \frac{TP_c}{TP_c + FN_c} \tag{16}$$

- **mean F1-score (mF1):** The harmonic mean of precision and recall, averaged across all classes to provide a comprehensive measure of segmentation quality:

$$\mathrm{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \mathrm{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{17}$$

$$\mathrm{F1}_c = \frac{2 \times \mathrm{Precision}_c \times \mathrm{Recall}_c}{\mathrm{Precision}_c + \mathrm{Recall}_c}, \quad \mathrm{mF1} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{F1}_c \tag{18}$$

3.3.2. Computational Cost Metrics

Considering the practical deployment requirements of lightweight segmentation models, the following computational cost and inference speed metrics are utilized:

- **Floating Point Operations (FLOPs):** The number of floating-point operations, expressed in GFLOPs (G), used to measure the theoretical computational complexity of the network.

- **Parameters (Params):** The total number of trainable parameters expressed in Millions (M), reflecting the memory footprint of the network.
- **Frames Per Second (FPS):** The number of images processed per second during inference. FPS is used as a practical reference for inference speed on the available GPU platform. FPS was measured on an NVIDIA RTX 4090 GPU using 512 × 512 input images with a batch size of 1. The first 50 iterations were discarded as warm-up, and the average inference time was calculated over 500 runs with AMP enabled.

3.4. Comparison with Other Networks

3.4.1. Quantitative Comparison

To evaluate the segmentation performance of LDS-Net, experiments were conducted under unified data splits, training strategies, and hyper-parameter settings. The proposed network was evaluated on two open datasets and compared with six representative semantic segmentation networks, including U-Net [52], DeepLabv3+ [29], HRNetv2 [53], SegFormer [54], DDRNet [55], and PIDNet-s [48]. The corresponding quantitative comparison results are summarized in Tables 1 and 2.

Table 1. Quantitative comparison of networks on USTD.

Model	mIoU (%)	mPA (%)	mF1 (%)	Acc (%)	IoU (%)	Recall (%)
U-Net [52]	79.46	86.05	87.85	94.37	65.22	74.44
DeepLabv3+ [29]	79.29	86.32	87.73	94.35	64.90	75.19
HRNetv2 [53]	77.14	84.69	86.19	93.67	68.09	72.25
SegFormer [54]	77.56	84.55	86.48	93.88	64.07	71.64
DDRNet [55]	79.97	87.88	88.22	94.39	66.24	78.85
PIDNet-s [48]	80.79	87.78	88.77	94.63	67.45	78.11
Ours	**81.70**	**89.93**	**89.42**	**94.76**	**69.30**	**82.70**

Note: The best results are shown in bold.

As summarized in Table 1, LDS-Net achieves the best performance across all evaluation metrics on USTD [50]. Specifically, the mIoU reaches 81.70%, outperforming the second-best network PIDNet-s [48] by 0.91%, while the IoU increases from 67.45% to 69.30% (+1.85%). Notably, LDS-Net yields a remarkable improvement in recall, jumping from 78.11% to 82.70% (+4.59%) compared to PIDNet-s [48], alongside a 2.15% boost in mPA. These substantial gains in recall and pixel accuracy indicate that LDS-Net reduces the false negative rate of hard-to-detect pixels. Consequently, it more effectively preserves the continuity of slender branch structures and mitigates the structural discontinuities commonly observed in conventional networks.

Table 2. Quantitative comparison of networks on N-ABSD.

Model	mIoU (%)	mPA (%)	mF1 (%)	Acc (%)	IoU (%)	Recall (%)
U-Net [52]	85.13	90.03	91.47	97.75	72.65	80.96
DeepLabv3+ [29]	84.32	89.59	90.93	97.62	71.15	80.18
HRNetv2 [53]	86.42	90.91	92.31	97.97	73.83	82.63
SegFormer [54]	82.20	87.38	89.49	97.30	70.70	75.76
DDRNet [55]	86.01	91.23	92.05	97.89	74.27	83.43
PIDNet-s [48]	86.74	91.39	92.45	98.00	75.39	83.65
Ours	**87.54**	**93.56**	**93.03**	**98.07**	**77.10**	**88.30**

Note: The best results are shown in bold.

As summarized in Table 2, LDS-Net also achieves the best performance on N-ABSD [51]. Compared to the second-best model, PIDNet-s [48], LDS-Net improves mIoU from 86.74% to 87.54% and mPA from 91.39% to 93.56%. The branch recall increases from

83.65% to 88.30%. These results suggest that LDS-Net is effective in complex orchard scenes, where leaf occlusion and background interference make slender branches difficult to detect.

3.4.2. Visual Comparison

To visually compare the differences between networks in segmenting slender branches, we selected two representative groups from USTD [50] and N-ABSD [51] respectively for visual comparison, denoted as USTD (a), USTD (b), N-ABSD (c), and N-ABSD (d), as illustrated in Figure 5. The white boxes highlight challenging regions involving fine branches, ambiguous boundaries, and severe occlusions, where the differences among the networks are most evident.

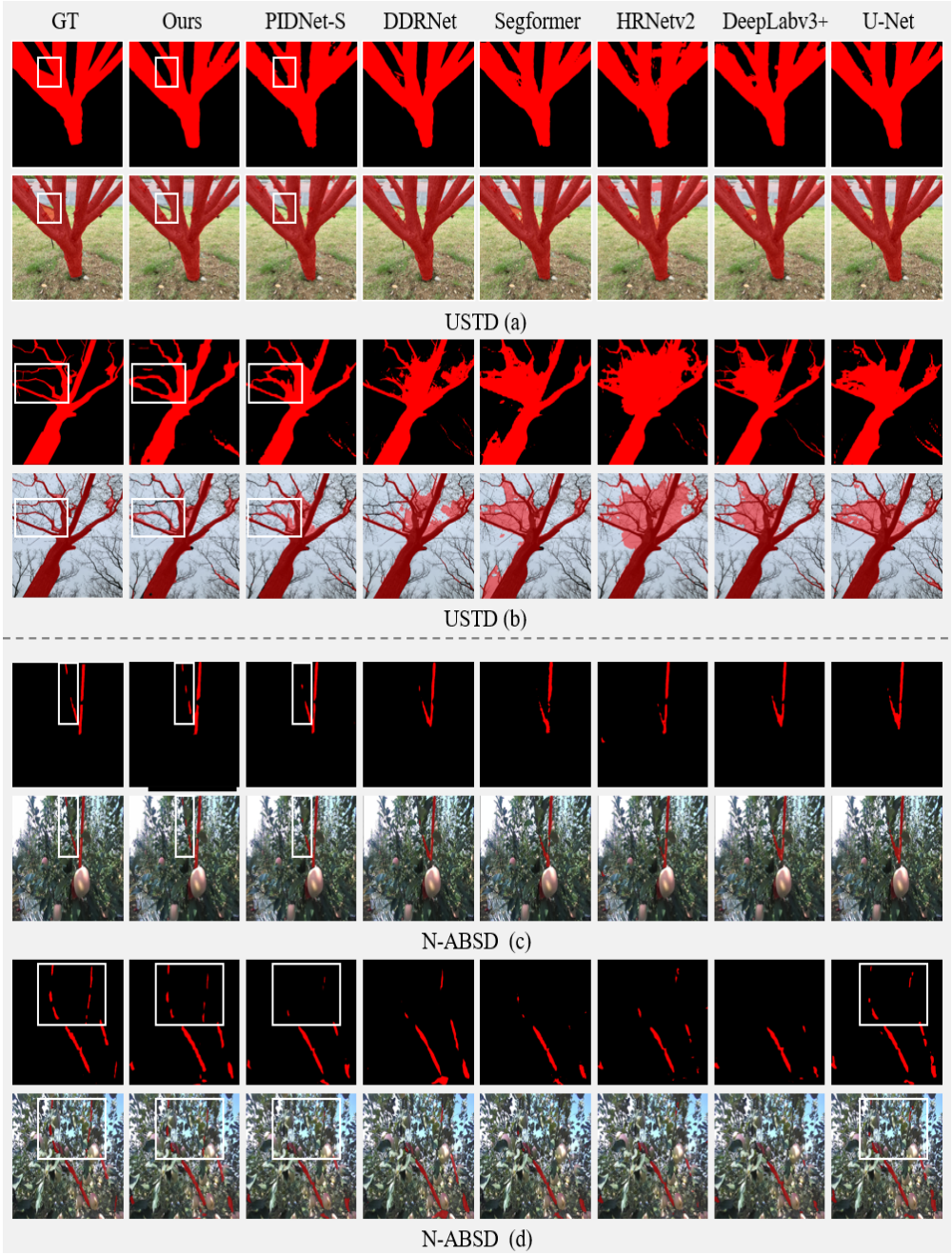


Figure 5. Qualitative comparison of branch segmentation results on USTD and N-ABSD. The red areas indicate the predicted branch mask and the white boxes highlight regions with fine branches, occlusions or blurred boundaries. Compared with other methods, LDS-Net preserves more complete branch structures and produces clearer boundaries.

- **USTD (a):** Within the white boxes, LDS-Net suppresses background noise, reduces non-branch responses, and produces clearer and more continuous boundaries than the other networks.
- **USTD (b):** In multi-fork slender branch regions under sky backgrounds, networks such as SegFormer [54] produce blob-like predictions with ambiguous boundaries, while PIDNet-s [48] preserves the overall branch orientation but misses some fine branch tips. In contrast, LDS-Net produces clearer and more accurate boundaries within the white boxes.
- **N-ABSD (c):** Within cluttered orchard scenes, the target branches are exceptionally fine. Most networks exhibit significant missed detections in the white boxes, whereas LDS-Net preserves more complete thin branch structures.
- **N-ABSD (d):** In scenes with severe occlusions caused by interlaced branches and leaves, DeepLabv3+ [29] and SegFormer [54] mainly detect coarse branches, while PIDNet-s [48] and U-Net [52] still suffer from discontinuous masks and missed detections. In contrast, LDS-Net maintains better structural integrity and branch connectivity, producing predictions that are more consistent with the ground-truth branch distribution.

The above visual comparison results are consistent with the quantitative comparison results reported in Tables 1 and 2.

3.4.3. Efficiency Comparison

Beyond segmentation accuracy, Table 3 compares the computational cost, parameter size, and practical inference speed of different networks. LDS-Net has the lowest FLOPs among all compared methods, with 14.19 G FLOPs and 11.95 M Params. Although its parameter count is slightly higher than that of PIDNet-s [48], LDS-Net reduces the computational cost by about 4.43 G FLOPs. In terms of practical inference speed, LDS-Net reaches 121.90 FPS on the available GPU platform. Although it is not the fastest model in FPS, it still achieves competitive practical inference speed. In particular, LDS-Net improves the branch IoU by about 1.85% compared with PIDNet-s on USTD (see Table 1). These results indicate that LDS-Net provides a favorable balance between segmentation accuracy, computational cost, and practical inference efficiency.

Table 3. Comparison of network computational cost and practical inference speed.

Model	FLOPs (G)	Params (M)	FPS (Frames/s)
U-Net [52]	225.84	24.89	115.39
DeepLabv3+ [29]	167.00	54.71	75.77
HRNetv2 [53]	90.93	29.54	40.57
SegFormer [54]	142.72	47.22	44.02
DDRNet [55]	17.98	20.15	**260.31**
PIDNet-s [48]	18.62	**7.62**	161.46
Ours	**14.19**	11.95	121.90

Note: The best results are shown in bold.

3.5. Ablation Study

To evaluate the effectiveness of the proposed modules and the multi-head compound loss function in LDS-Net, ablation experiments were conducted on two open datasets: USTD [50] and N-ABSD [51]. These experiments focus on module effectiveness and the multi-head compound loss.

3.5.1. Module Effectiveness Analysis

Tables 4 and 5 present the step-by-step ablation results on USTD and N-ABSD. The baseline model adopts ResNet18 [46] as the backbone and uses the Simply-C module for feature fusion. In Simply-C, features from the B_{context} and BG are first resized to the same spatial resolution and concatenated along the channel dimension. The concatenated features are then fused by two consecutive 3×3 convolution layers with batch normalization and ReLU activation.

- **Effect of the dual-branch backbone:**

Replacing ResNet18 [46] with the proposed dual-branch backbone (B1) brings consistent improvements on both datasets. Compared with the Base model, B1 improves mIoU and mF1 by 0.84% and 0.52% on USTD, and by 0.42% and 0.27% on N-ABSD. Meanwhile, the computational cost is clearly reduced, with FLOPs decreasing by 8.65 G and Params by 6.03 M.

This shows that the proposed backbone is more accurate and lighter than the ResNet18-based baseline. The improvement mainly comes from the division of roles between two branches. The Detail-Aware Branch focuses on local slender structures and branch continuity, while the Context-Aware Branch captures broader contextual information. The reduction in complexity is mainly due to the lightweight DSHRB and MERB designs, which replace the heavier ResNet18 backbone. Therefore, B1 provides a better accuracy–complexity balance for thin and curvilinear branch segmentation.

- **Effect of the GTT module:**

Introducing GTT while retaining the Simply-C module for feature fusion (B2) further improves segmentation performance. Compared with B1, B2 improves mIoU by 0.54% on USTD and 0.52% on N-ABSD. It also improves mF1 by 0.31% on USTD and 0.33% on N-ABSD. These improvements indicate that GTT helps the network model structural relationships in complex scenes. By using W-MSA and OP on high-level features, GTT provides stronger structural cues for recovering occluded or fragmented branches.

The additional complexity of GTT mainly comes from W-MSA, OP, and FFN. W-MSA introduces Q, K, and V projections and window-based attention computation. OP adds convolution operations for offset prediction, and FFN adds channel-wise feature interaction. Compared with B1, B2 increases FLOPs from 13.11 G to 14.04 G and Params from 10.95 M to 11.80 M. This additional cost is moderate and brings stable accuracy gains on both datasets.

- **Effect of the PAF module:**

Replacing the Simply-C module with the proposed PAF without GTT (B3) brings limited but positive improvements. Compared with B1, B3 improves mIoU by 0.18% on USTD and 0.47% on N-ABSD. The improvement is small because PAF is mainly a feature fusion module. It can reweight detail, context, and boundary-guided features, but it does not directly introduce stronger high-level structural information. Therefore, without GTT, PAF has limited ability to recover occluded or fragmented branch structures.

The computational overhead of PAF is small because it only uses a lightweight attention generator and a compact fusion sub-network. Compared with B1, B3 only increases FLOPs from 13.11 G to 13.26 G and Params from 10.95 M to 11.10 M.

When GTT and PAF are used together, the performance improves more clearly. Compared with B2, the full model only adds 0.15 G FLOPs and 0.15 M Params, but improves mIoU by 0.41% on USTD and 0.74% on N-ABSD. This indicates that PAF can better use the structural information enhanced by GTT and fuse it with detail and boundary-guided cues.

Overall, compared with B1, the full model increases FLOPs by 1.08 G and Params by 1.00 M, while improving mIoU by 0.95% on USTD and 1.26% on N-ABSD. These results

show that GTT provides stronger structural modeling, while PAF adds only a small extra cost for adaptive fusion. Therefore, the full model achieves a better balance between segmentation accuracy and computational cost.

Table 4. Ablation experiments on USTD.

Model	Backbone		Global	Fusion		mIoU	mF1	Params	FLOPs
	B_{detail}	B_{context}	GTT	Simply-C	PAF	(%)	(%)	(M)	(G)
Base	R-18	R-18	×	✓	×	79.91	88.23	16.98	21.76
B1	DSHRB	MERB	×	✓	×	80.75	88.75	**10.95**	**13.11**
B2	DSHRB	MERB	✓	✓	×	81.29	89.06	11.80	14.04
B3	DSHRB	MERB	×	×	✓	80.93	88.87	11.10	13.26
Ours	DSHRB	MERB	✓	×	✓	**81.70**	**89.42**	11.95	14.19

Note: The best results are shown in bold.

Table 5. Ablation experiments on N-ABSD.

Model	Backbone		Global	Fusion		mIoU	mF1	Params	FLOPs
	B_{detail}	B_{context}	GTT	Simply-C	PAF	(%)	(%)	(M)	(G)
Base	R-18	R-18	×	✓	×	85.86	91.96	16.98	21.76
B1	DSHRB	MERB	×	✓	×	86.28	92.23	**10.95**	**13.11**
B2	DSHRB	MERB	✓	✓	×	86.80	92.56	11.80	14.04
B3	DSHRB	MERB	×	×	✓	86.75	92.53	11.10	13.26
Ours	DSHRB	MERB	✓	×	✓	**87.54**	**93.03**	11.95	14.19

Note: The best results are shown in bold.

3.5.2. Module Visualization Analysis

Figure 6 provides a visual illustration of the segmentation results obtained after progressively stacking different modules. We selected two representative groups from USTD [50] and N-ABSD [51] respectively for visual comparison, denoted as USTD (a), USTD (b), N-ABSD (c), and N-ABSD (d).

The rightmost columns display the local overlay of the segmentation masks on the original images corresponding to the white boxes, providing a more intuitive illustration of the effect of each module. These white boxes highlight challenging regions involving fine branches, ambiguous boundaries, and occlusions, where the performance differences among the configurations are most evident.

The results indicate that the baseline model exhibits frequent missed detections and structural discontinuities when segmenting slender branches. Introducing the proposed dual-branch backbone (B1) noticeably improves branch continuity, although some boundary ambiguity remains. With the employment of GTT (B2), robustness to occlusion is significantly enhanced, enabling more reliable inference of partially occluded branch structures. Furthermore, after applying PAF (Ours), fine-grained branches are more completely preserved, and branch boundaries become sharper and better separated from the background.

The above visual results are consistent with the quantitative results reported in Tables 4 and 5.

3.5.3. Multi-Head Compound Loss Function Analysis

Table 6 analyzes the contributions of different components in the proposed multi-head compound loss function. When only the prediction output of $\text{Main}_{\text{head}}$ is supervised using $\mathcal{L}_{\text{focal}}$ (S1), the network achieves mIoU scores of 79.84% on USTD and 84.74% on N-ABSD, indicating that relying only on $\mathcal{L}_{\text{focal}}$ is insufficient to capture the spatial continuity of slender branch structures.

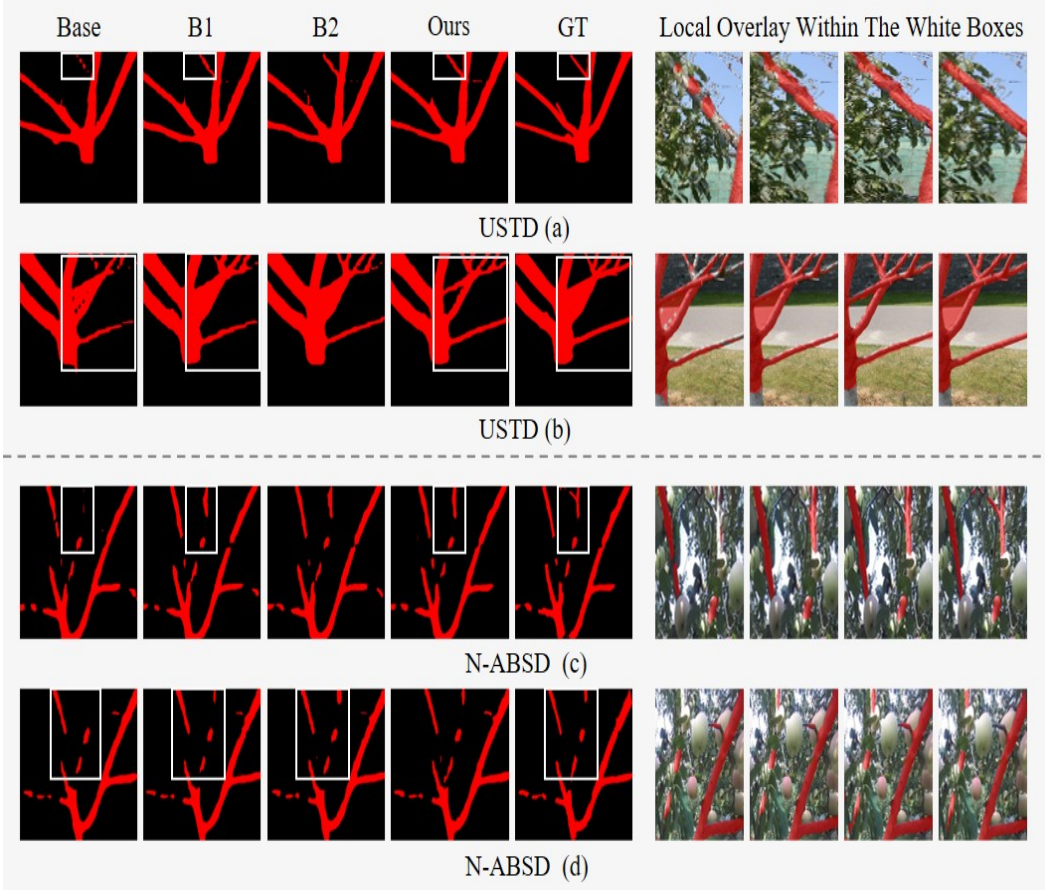


Figure 6. Ablation visualization on USTD and N-ABSD. From left to right: segmentation masks produced by the Base model, B1 (dual-branch backbone), B2 (+GTT), Ours (+PAF), and the ground truth (GT). The rightmost panels show local overlays of the predicted masks in the regions marked by white boxes. Red areas denote predicted branch masks.

Table 6. Ablation study of different loss components.

Method	$\text{Main}_{\text{head}}$				Auxiliary Heads	USTD	N-ABSD
	$\mathcal{L}_{\text{focal}}$	$\mathcal{L}_{\text{dice}}$	$\mathcal{L}_{\text{dice}}^{\text{ms}}$	$\mathcal{L}_{\text{bc}}$		mIoU (%)	mIoU (%)
S1	✓	×	×	×	×	79.84	84.74
S2	✓	✓	×	×	×	81.08	87.26
S3	✓	✓	×	✓	×	81.21	87.48
S4	✓	×	✓	✓	✓	**81.70**	**87.54**

Note: The best results are shown in bold.

After incorporating $\mathcal{L}_{\text{dice}}$ (S2), the mIoU improves to 81.08% on USTD and 87.26% on N-ABSD, respectively, validating the effectiveness of Dice Loss in enhancing regional consistency and structural completeness.

Further employing Boundary-Weighted Cross-Entropy Loss $\mathcal{L}_{\text{bc}}$ (S3) leads to continued performance gains (81.21% on USTD, 87.48% on N-ABSD), confirming the effectiveness of explicit boundary supervision, especially under severe occlusion scenes in N-ABSD.

Finally, S4 achieves the best performance when the detail-aware and boundary-guided auxiliary heads are jointly supervised and the multi-head Dice Loss is used for the main prediction. This setting reaches 81.70% mIoU on USTD and 87.54% mIoU on N-ABSD.

4. Conclusions

The precise segmentation of slender tree branches in complex natural environments is a critical prerequisite for advanced agricultural and forestry applications. The experimental

results obtained on USTD [50] and N-ABSD [51] datasets show that the proposed LDS-Net can better handle severe occlusions, structural discontinuities, and ambiguous boundaries. Compared with representative semantic segmentation networks, LDS-Net achieves better segmentation accuracy with a low computational cost.

Our findings highlight the importance of separating detail preservation from contextual modeling. Previous lightweight architectures, such as PIDNet-s [48] and DDRNet [55], can achieve high inference speed but may lose some fine structural details of thin targets. This is reflected by their lower branch recall values such as 78.11% for PIDNet-s [48] on USTD. By introducing Dynamic Snake Convolution (DSConv) [49] into the Detail-Aware Branch, LDS-Net can adapt its receptive field to the curvilinear trajectories of branches. This design helps improve the detection of slender branch regions, as reflected by the branch recall values of 82.70% on USTD and 88.30% on N-ABSD. These results suggest that dynamic local feature extraction is more suitable for slender branch structures than fixed rectangular convolutions.

Furthermore, dense foliage and variable illumination in natural scenes often lead to severe occlusions. Transformer-based models such as SegFormer [54] have strong global modeling capability, but their high computational cost and blob-like predictions limit their performance in fine-grained branch segmentation. The integration of GTT and PAF in LDS-Net helps address this issue. Visual comparisons show that LDS-Net can better preserve partially occluded branch structures while maintaining clearer boundaries.

For practical agricultural and ecological monitoring, both segmentation accuracy and inference efficiency are important. LDS-Net achieves the highest mIoU while requiring only 14.19 G FLOPs, representing a 24% reduction in computational complexity compared to PIDNet-s [48]. It also reaches 121.90 FPS on the available GPU platform. These results suggest the potential of LDS-Net for future deployment on resource-constrained agricultural platforms, such as field monitoring systems and pruning-related applications.

Despite these results, some limitations remain. As shown in Figure 7, LDS-Net may still produce imperfect predictions in extremely challenging cases. In USTD, densely distributed thin branches and complex backgrounds make it difficult to recover all fine branch structures, leading to missed or discontinuous predictions. In N-ABSD, occlusion caused by dense leaves and fruits may obscure local branch regions. These cases indicate that, although LDS-Net can preserve much of the main branch topology, its prediction continuity is still limited under severe occlusion and highly dense thin branch structures.

The failure cases in Figure 7 also reflect common difficulties in thin-structure segmentation. Recent studies have addressed similar problems from different perspectives. CrackSegFlow [56] focuses on crack segmentation and uses topology-preserving mask injection and boundary-gated modulation during data synthesis to maintain thin-structure continuity. Zhu et al. [57] proposed a prompt-free curvilinear structure segmentation framework, where internal adapters refine local curvilinear representations and external adapters support cross-domain feature alignment. LiteRoadSegNet [58] focuses on lightweight road extraction and uses topology-aware supervision to improve road connectivity under limited computational cost. Together, these studies suggest that local detail sensitivity, boundary-aware modeling, and continuity preservation are important for thin-structure segmentation. In this context, LDS-Net can be regarded as a branch-oriented lightweight design that combines detail preservation, contextual modeling, boundary guidance, and feature fusion. These designs may provide useful insights for other thin-structure segmentation tasks, such as roots, vines, roads, vessels, cracks, and power lines, because these targets also require accurate local boundaries and continuous structural prediction.

The current experiments are limited to two branch segmentation datasets. Other thin-structure tasks may have different imaging modalities, object scales, texture patterns,

background complexity, annotation styles, and topology requirements. For example, roots and vines may share natural-scene occlusion with branches, but their growth patterns and backgrounds can be different. Roads and power lines are often captured in remote sensing images and may show larger scale changes. Vessels and cracks may appear in medical or inspection images with different contrast and noise characteristics. Therefore, LDS-Net may require retraining, fine-tuning, or task-specific modification before being applied to these domains.

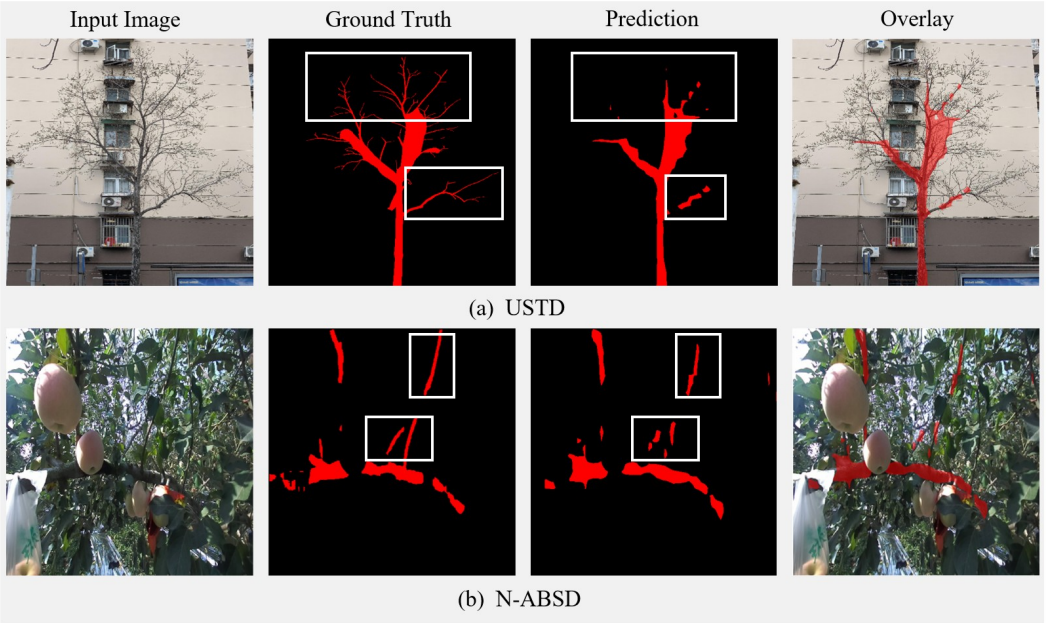


Figure 7. Representative failure cases of LDS-Net. White boxes highlight missed or discontinuous branch regions, and red regions indicate branch masks.

To mitigate these limitations, several directions will be explored in future work. First, topology-aware constraints, skeleton supervision, and centerline-based losses will be considered to improve the continuity of slender branch predictions. These methods can encourage the network to preserve connected and centerline-consistent structures, which may help reduce fragmented predictions of partially visible branches. Second, lightweight multimodal fusion with RGB-D images or sparse point clouds may provide additional spatial cues when 2D RGB information is insufficient. This may improve model robustness under severe occlusion. Third, LDS-Net will be evaluated on more thin-structure datasets to examine its cross-domain generalization. For tasks with different appearances, scales, imaging modalities, or topology requirements, task-specific fine-tuning or domain adaptation may be needed.

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Data Availability Statement: The datasets used in this study are publicly available. The Urban Street Tree Dataset (USTD) [50] can be accessed at https://ytt917251944.github.io/dataset_jekyll (accessed on 8 July 2026), and the NWPU-Apple Branch Segmentation Dataset (N-ABSD) [51] is available at https://github.com/ly27253/LED-Net (accessed on 8 July 2026).

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